  - Ensure compliance with relevant regulations (e.g., GDPR, HIPAA).
  - Encrypt sensitive data at rest and in transit.
  - Anonymize or pseudonymize data where possible.
- Security
  - Implement secure coding practices.
  - Protect against cyberattacks (e.g., rate limiting, firewalls).
  - Regularly audit API key usage.
  - Rotate API keys periodically.
  - Never commit secrets to version control.
- Performance and latency
  - Assess the effect of network latency and mitigate delays.
  - Implement local caching where applicable.
  - Optimize API calls and data retrieval for efficiency.
- Scalability and reliability
  - Implement CI/CD pipelines for automated deployment.
  - Ensure high availability with redundancy and failover strategies .
- Monitoring and logging
  - Set up real-time monitoring and alerting.
  - Implement detailed logging for debugging and auditing.
- Compliance
  - Understand and adhere to data policies.
  - Ensure compliance with industry regulations and security standards.
- Deployment strategy
  - Choose the right deployment platform (e.g., AWS, GCP, Azure).
  - Use deployment strategies to minimize downtime (e.g., blue-green, canary, rolling updates).

Many of the checklist items mentioned have been discussed in the current and previous chapters, so you may find some familiar concepts. However, there are certain aspects that we have not covered, which primarily fall outside the scope of this book.

9.6.3 *Example deployment to Vercel*

Let's use our deployment checklist to deploy our simple chat application with the security controls we've discussed.

> **NOTE** The source code for the example in this chapter is available at https://github.com/Generative-AI-Web-Apps/Code/tree/main/ch09/chat-deployment.